CHEM 427/527 · SESSION 12 · TUESDAY, NOVEMBER 10, 2026

Peak detection and feature extraction: does this lavender oil meet the pharmacopoeia?

Matt Bush, with teaching assistants Lucas Narisawa and Chris Weir

Drafted with assistance by Claude Code. Reviewed and revised by Matt.

LEARNING OBJECTIVES

By the end of today you can

1. Choose peak-detection settings from a chromatogram’s noise, its narrowest peak, and the smallest peak that must count, and show what a threshold set too low or too high does.
2. Integrate each peak between the lowest points that separate it from its neighbors, and report each component as a percent of the summed area of the peaks counted.
3. Decide whether an oil meets a specification from three replicate runs, their spread, and the distance to the nearest limit.
4. Integrate the same peaks by a second rule, show how far it moves each result, and say whether it changes your call.
5. Write a skill for a step you repeat, check it against a table you built by hand, and decide whether to change it before it reads new data.

FIRST, EVERYONE PULLS

Get session 12 with git pull

```
git pull
```

S12-peak-detection appears, with

- **data/L13A.cdf**: Dataset A, the worked example
- three runs of each oil, such as **data/L19A.cdf**, **L19B.cdf** and **L19C.cdf**
- **data/samples.csv**: each student number's oil
- **data/README.md**: where the runs come from and how to read them

Session 11 is due today

- Upload it by 1:30 pm.
- Fill in your prompt record, then Restart and Run All before you upload.

Nothing is downloaded today: the 91 runs and samples.csv are in the repository.

THE DECISION

Does this lavender oil meet the pharmacopoeia?

Who decides

A quality-control analyst at a supplier or a pharmacy reads a GC-FID run against the European Pharmacopoeia's profile, by area percent: camphor at most 1.2%, linalool 20.0 to 45.0%, and linalyl acetate 25.0 to 47.0%.

Passed wrongly

An oil cut with lavandin or with synthetic linalool is sold as pharmacopoeial.

Failed wrongly

A genuine lot is rejected, and its supplier loses the sale.

Real data: 55 commercial oils, three GC-FID runs each

Dataset A, L13A.cdf, is a 73 min run on a wax column, one point every 0.1 s. The solvent peaks at 4.39 min and 1,413 mV. Linalool gives the largest of the oil's peaks, 124 mV on a baseline near 1 mV. The files name no peak; one oil's GC-MS run confirmed camphor's place. You have one oil's three runs.

FROM WHERE THE BASELINE SITS TO WHICH PEAKS COUNT

What does integrating every peak down to 0 mV do to linalool’s area percent?

Session 11

An impurity’s area percent depended on where the baseline was drawn.

Today

The baseline sits near 1 mV, under dozens of peaks a few mV tall.

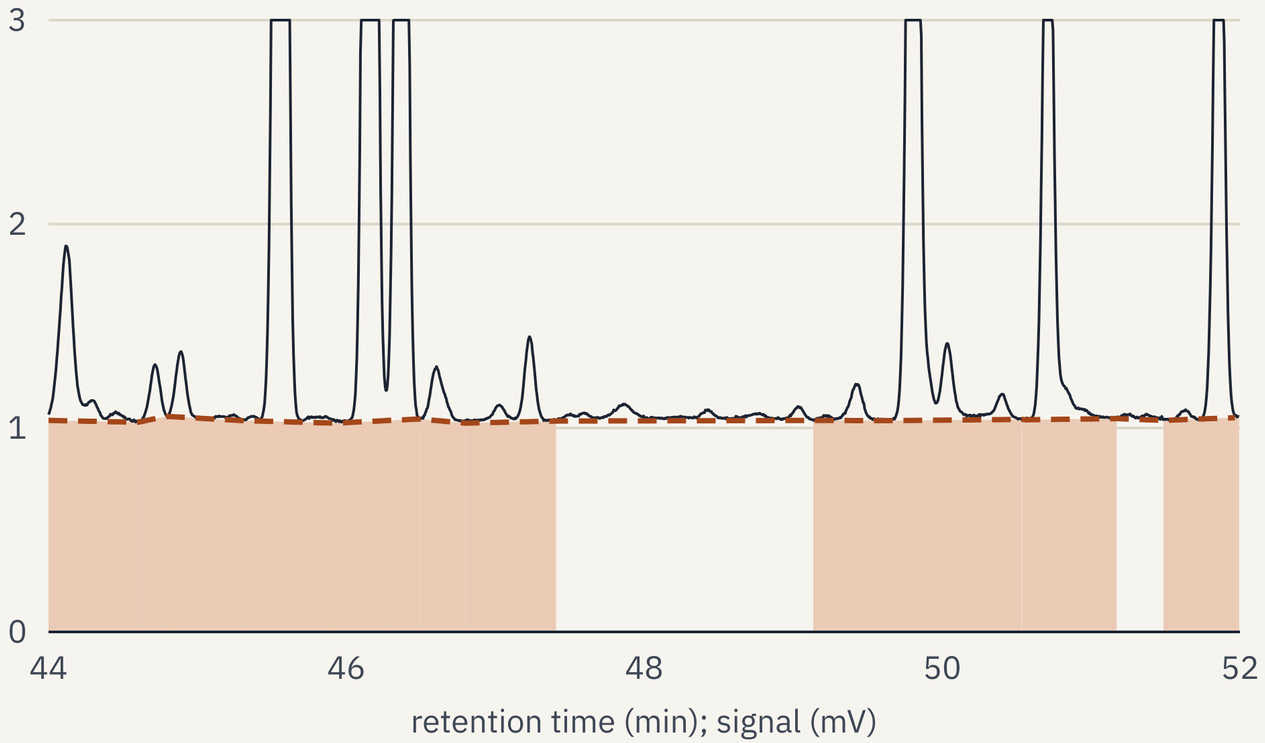
The question

Each peak is integrated down to 0 mV instead of down to that baseline. Where does linalool’s share go?

Does linalool’s area percent go up, go down, or stay the same?

THE ANSWER, ON DATASET A

Each peak gains the slab under it



The drop-line baseline, near 1 mV

The slab that 0 mV adds

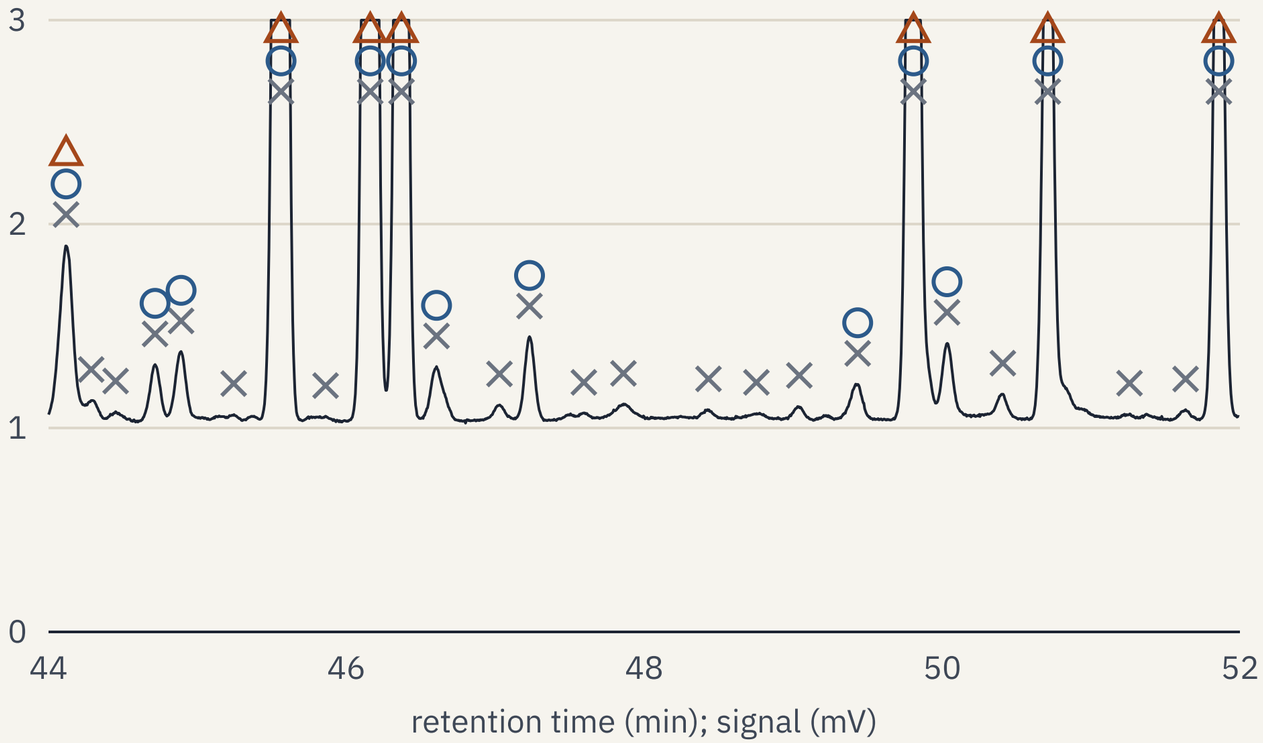
Integrated to	baseline	0 mV
camphor	0.25	0.80
linalool	34.37	17.79
linalyl acetate	35.06	18.35

The table gives each component’s area percent. The total grows faster than linalool’s area, so linalool falls below its 20.0% limit and camphor rises.

A skill that does not say where the baseline is will choose one.

LIVE: THREE PROMINENCES ON DATASET A

Which peaks count toward the total?



- **5 times the noise, 0.024 mV**: 122 found, 54 kept
- **The disregard-limit setting, 0.124 mV**: 58 found, 53 kept
- **Read off the plot, 0.5 mV**: 32 found, missing 21 that the disregard-limit setting keeps

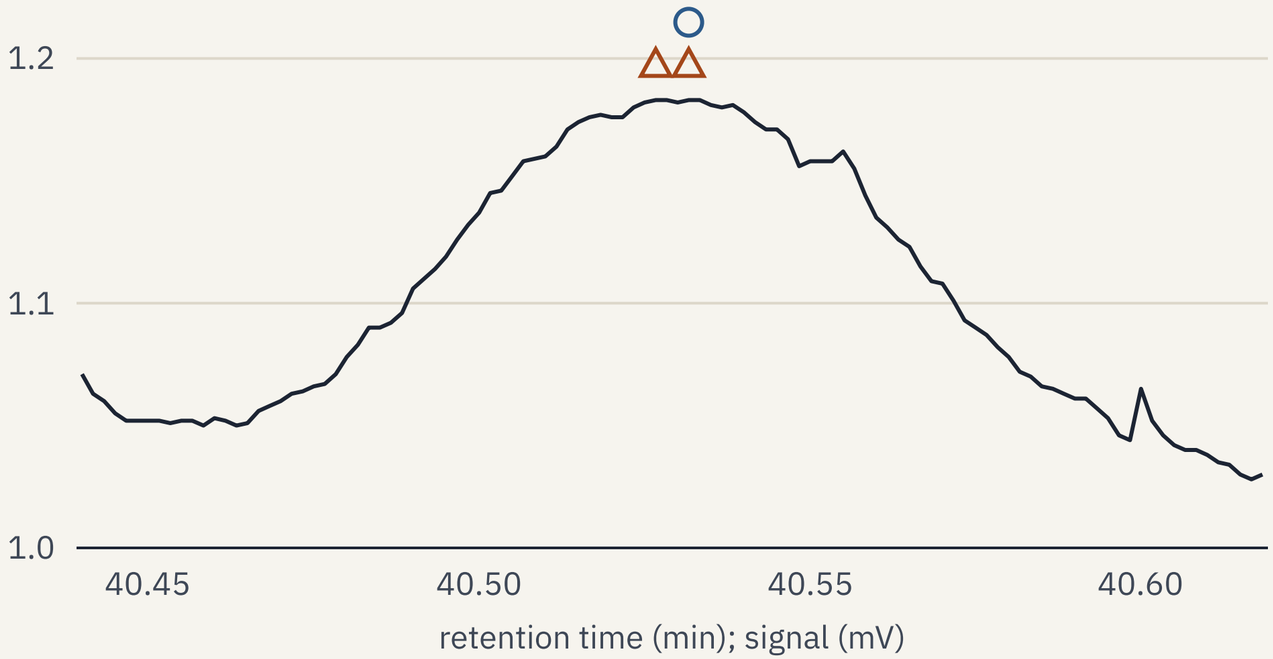
A peak is kept at or above 0.05% of the total area, the disregard limit. From 44 to 52 min the chart marks 26 crosses, 13 circles and 7 triangles.

The data system’s own table, inside the file, lists 38 peaks.

Too low counts noise. Too high misses real peaks.

LIVE: THE SAME DETECTION WITH NO DISTANCE

What keeps one apex per peak?



△ **No distance**: 60 peaks; the peak shown is found twice, at 40.527 and 40.532 min, 0.3 s apart

○ **Distance 35 points**: one apex

The distance is the fewest points allowed between two apexes.

Section 1 sets every setting from two numbers in the run: its noise and its narrowest peak. Your oil's numbers differ.

From what number in the data does the notebook set the distance?

ASSIGNMENTS

One oil each, by student number

#	Oil	#	Oil	#	Oil	#	Oil
1	L6	8	L51	15	L34	21	L19
2	L31	9	L50	16	L25	22	L36
3	L37	10	L3	17	L38	23	L12
4	L44	11	L18	18	L16	24	L29
5	L55	12	L11	19	L43	25	L23
6	L47	13	L1	20	L14	26	L21
7	L39	14	L9				

Your oil is beside your student number from session 4 in data/samples.csv. In Parameters, set **SAMPLE**, such as SAMPLE = "L19", and run the Parameters cell. Section 1 is the worked example on Dataset A. Section 2 is your oil, by hand and by the skill, and section 3 compares the two integration rules.

AI PRACTICE 7 OF 8, NEW TODAY: A SKILL.MD FOR A REPEATED STEP

A file of instructions, run as a command

Live, on Dataset A, in VS Code opened on the repository’s folder

1. In the Explorer, make .claude/skills/peak-report/SKILL.md and paste the template from section 2.

2. Fill the four blanks with section 1’s values: 0.124 mV, 17.6 points, 35 points and drop. Save.

3. In the Claude Code prompt box, type / and choose Clear conversation. Type /skills and find peak-report in the list.

4. Type the command, with Dataset A’s path:

```
/peak-report sessions/S12-peak-detection/data/L13A.cdf
```

What the template states as numbers

- The stretch searched, 5.8 to 69.5 min.
- Every window, in points: 300 for a bound’s search, 180 for the baseline test.
- How each bound is searched.
- Where the baseline is.

A wording that the assistant can read two ways is a step to change.

THE CHECK, ON DATA WHOSE ANSWER YOU KNOW

Does the skill reproduce your hand-built table?

Dataset A, area percent, drop-line at 0.124 mV

Component	By hand	The skill	Data system
camphor	0.25	?	0.25
linalool	34.37	?	35.16
linalyl acetate	35.06	?	35.86

The data system’s column comes from its own list of 38 peaks. It is a third reading, not the answer.

How to read the report

- Its first line: the parameters and the rule it used.
- Its three values against yours, within 0.01 points.
- Anything it took from outside SKILL.md.

A match counts only if SKILL.md’s own text made it. A larger difference points at a step to change before the skill reads your runs.

SECTION 2: THE SKILL ON YOUR RUNS, THEN TASK 3

Your settings, your table, your record

Known answer first, new data second

1. Check the skill on Dataset A. Your prompt is the record’s first line.
2. TASK 2: your settings, from run A’s printout.
3. Put your three numbers in SKILL.md, save, and run /peak-report on runs A, B and C.
4. TASK 3: your own table, in a code cell you add.
5. Fill the record’s other two lines.

The record, a markdown cell

```
**Prompt record**
- **My prompt:**
- **The skill's Dataset A values against my hand-built ones, and my runs' against TASK 3's:**
- **What I changed in SKILL.md, or what I verified (the report's parameter line, and anything it took from outside SKILL.md):**
```

TASK 3: a code cell you write that sets **hand**, a table with one row per run, A, B and C, and a column each for camphor, linalool and linalyl acetate.

The three lines of your prompt record count toward execution.

THE JUDGMENT CELL, SECTION 4

Your oil’s call, and one entry

(a) Your answer

- Does your oil meet the profile for camphor, linalool and linalyl acetate?
- Each component’s mean and spread, the nearest limit and your margin, the other rule’s shift, and the data system’s value
- What a wrong call costs either way

(b) One judgment entry

- Your prompt record is the source.
- What you did: changed, rejected, or verified
- The evidence: the skill’s Dataset A values against your hand-built ones

Which of the three distances, the margin, the spread or the rule’s shift, would make you ask for a fourth run of your oil?

SECTION 1 IS READ AND RUN. THE SKILL AND SECTION 2 ARE YOURS.

Go, then submit

Submit

Fill in your prompt record, then Restart, Run All, save, and upload activity.ipynb by **1:30 pm Thursday, November 12**.

Scoring, 0 to 3

1 if it runs after a restart and holds your prompt record, **0** if not.

0 to 2 for the judgment cell.

Help

Ask Matt, Lucas, or Chris in the room.

The third hour is an office hour.